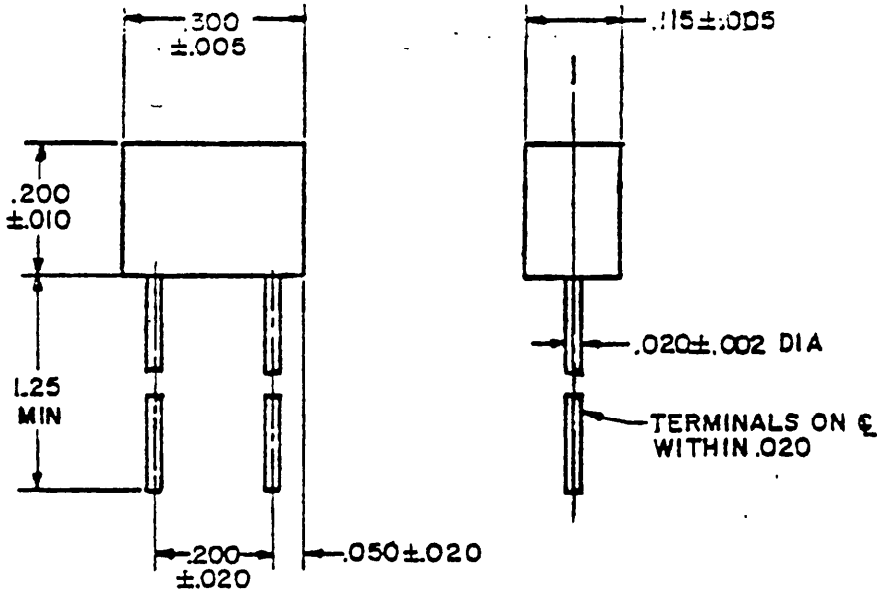
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC,

STYLES CY06 AND CY06.....TX

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11272.

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-C-23269.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.200	5.08
.005	.13	.050	1.27	.300	7.62
.010	.25	.115	2.92	1.25	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch - 25.4 mm.

FIGURE 1. Capacitors, Fixed, Glass Dielectric, Styles CY06 and CY06.... TX.

FSC 5910

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REQUIREMENTS:

Design and construction:
Dimensions and configuration – See figure.
Material:
Dielectric – Glass.
Case – Epoxy.
Lead – Gold plated dumet (MIL-STD-1276, type D).

Capacitance value – See table I.

Capacitance tolerance – See table I.

DC rated voltage – 300 volts.

Temperature range – -55° to +125°C.
Dielectric withstanding voltage: Method 301 of MIL-STD-202 – 200 percent dc rated voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure: Method 105 of MIL-STD-202.
Test condition B (50,000 ft).
150 percent dc rated voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance: Method 302 of MIL-STD-202.
At 25°C – 100,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor:
Not to exceed 0.002 for capacitance values of 1 to 100 pF.
Not to exceed 0.001 for capacitance values greater than 100 pF.

Quality factor (Q):
Method 306 of MIL-STD-202.
Test frequency – 1 MHz ±200 kHz.
Not less than 500 for capacitance values of 1 to 100 pF.
Not less than 1,000 for capacitance values greater than 100 pF.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I (100 G).

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B (15G).

Temperature cycling: Method 102 of MIL-STD-202, test condition C.

Immersion: Method 104 of MIL-STD-202, test condition B.
Dielectric withstanding voltage – No breakdown.
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value or 0.5 pF, whichever is greater, from initial value.
Dissipation factor – Not to exceed 0.002 for capacitance values of 1 to 100 pF. Not to exceed 0.001 for capacitance values greater than 100 pF.

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TABLE I. *Styles CY06 and CY06.....TX.*

Type designation 1/	Capacitance pF	DC rated voltage	Capacitance tolerances available (±)	Type designation 1/	Capacitance pF	DC rated voltage	Capacitance tolerances available (±)
CY06C1R0C	1	300	C	CY06C150-TX	15	300	CGJ
CY06C1R0CTX	1	300	C	CY06C160-	16	300	CGJ
CY06C1R5C	1.5	300	C	CY06C160-TX	16	300	CGJ
CY06C1R5CTX	1.5	300	C	CY06C180-	18	300	CGJ
CY06C2R2-	2.2	300	CD	CY06C180-TX	18	300	CGJ
CY06C2R2-TX	2.2	300	CD	CY06C200-	20	300	CGJ
CY06C2R7C	2.7	300	C	CY06C200-TX	20	300	CGJ
CY06C2R7CTX	2.7	300	C	CY06C220-	22	300	CGJ
CY06C3R0-	3.0	300	CD	CY06C220-TX	22	300	CGJ
CY06C3R0-TX	3.0	300	CD	CY06C240-	24	300	CGJ
CY06C3R3C	3.3	300	C	CY06C240-TX	24	300	CGJ
CY06C3R3CTX	3.3	300	C	CY06C270-	27	300	FGJ
CY06C3R6-	3.6	300	CD	CY06C270-TX	27	300	FGJ
CY06C3R6-TX	3.6	300	CD	CY06C300-	30	300	FGJ
CY06C3R9C	3.9	300	C	CY06C300-TX	30	300	FGJ
CY06C3R9CTX	3.9	300	C	CY06C330-	33	300	FGJ
CY06C4R3-	4.3	300	CD	CY06C330-TX	33	300	FGJ
CY06C4R3-TX	4.3	300	CD	CY06C360-	36	300	FGJ
CY06C4R7C	4.7	300	C	CY06C360-TX	36	300	FGJ
CY06C4R7CTX	4.7	300	C	CY06C390-	39	300	FGJ
CY06C5R1-	5.1	300	CJ	CY06C390-TX	39	300	FGJ
CY06C5R1-TX	5.1	300	CJ	CY06C430-	43	300	FGJ
CY06C5R6-	5.6	300	CJ	CY06C430-TX	43	300	FGJ
CY06C5R6-TX	5.6	300	CJ	CY06C470-	47	300	FGJ
CY06C6R2-	6.2	300	CJ	CY06C470-TX	47	300	FGJ
CY06C6R2-TX	6.2	300	CJ	CY06C510-	51	300	FGJ
CY06C6R8-	6.8	300	CJ	CY06C510-TX	51	300	FGJ
CY06C6R8-TX	6.8	300	CJ	CY06C560-	56	300	FGJ
CY06C7R5-	7.5	300	CJ	CY06C560-TX	56	300	FGJ
CY06C7R5-TX	7.5	300	CJ	CY06C620-	62	300	FGJ
CY06C8R2-	8.2	300	CJ	CY06C620-TX	62	300	FGJ
CY06C8R2-TX	8.2	300	CJ	CY06C680-	68	300	FGJ
CY06C9R1-	9.1	300	CJ	CY06C680-TX	68	300	FGJ
CY06C9R1-TX	9.1	300	CJ	CY06C750-	75	300	FGJ
CY06C100-	10	300	CJ	CY06C750-TX	75	300	FGJ
CY06C100-TX	10	300	CJ	CY06C820-	82	300	FGJ
CY06C110-	11	300	CJ	CY06C820-TX	82	300	FGJ
CY06C110-TX	11	300	CJ	CY06C910-	91	300	FGJ
CY06C120-	12	300	CJ	CY06C910-TX	91	300	FGJ
CY06C120-TX	12	300	CJ	CY06C101-	100	300	FGJ
CY06C130-	13	300	CGJ	CY06C101-TX	100	300	FGJ
CY06C130-TX	13	300	CGJ	CY06C111-	110	300	FGJ
CY06C150-	15	300	CGJ	CY06C111-TX	110	300	FGJ

1/ See footnote at end of table.

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TABLE I. *Styles CY06 and CY06.....TX – Continued.*

Type designation 1/	Capacitance pF	DC rated voltage	Capacitance tolerances available (±)	Type designation 1/	Capacitance pF	DC rated voltage	Capacitance tolerances available (±)
CY06C121-	120	300	FGJ	CY06C271-TX	270	300	FGJ
CY06C121-TX	120	300	FGJ	CY06C301-	300	300	FGJ
CY06C131-	130	300	FGJ	CY06C301-TX	300	300	FGJ
CY06C131-TX	130	300	FGJ	CY06C331-	330	300	FGJ
CY06C151-	150	300	FGJ	CY06C331-TX	330	300	FGJ
CY06C151-TX	150	300	FGJ	CY06C361-	360	300	FGJ
CY06C161-	160	300	FGJ	CY06C361-TX	360	300	FGJ
CY06C161-TX	160	300	FGJ	CY06C391-	390	300	FGJ
CY06C181-	180	300	FGJ	CY06C391-TX	390	300	FGJ
CY06C181-TX	180	300	FGJ	CY06C431-	430	300	FGJ
CY06C201-	200	300	FGJ	CY06C431-TX	430	300	FGJ
CY06C201-TX	200	300	FGJ	CY06C471-	470	300	FGJ
CY06C221-	220	300	FGJ	CY06C471-TX	470	300	FGJ
CY06C221-TX	220	300	FGJ	CY06C511-	510	300	FGJ
CY06C241-	240	300	FGJ	CY06C511-TX	510	300	FGJ
CY06C241-TX	240	300	FGJ	CY06C561-	560	300	FGJ
CY06C271-	270	300	FGJ	CY06C561-TX	560	300	FGJ

1/ Complete type designation will include an additional letter symbol to indicate capacitance tolerance, where applicable, and the letters TX for TX version parts.

Examples: Non-TX version – CY06C1ROC
TX version – CY06C1R0CTX

Moisture resistance: Method 106 of MIL-STD-202.
At 90 to 95 percent relative humidity:
Insulation resistance – Not less than 1,000 megohms.
At 50 ±5 percent relative humidity:
Dielectric withstanding voltage – No breakdown.
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value or 0.5 pF, whichever is greater.
Dissipation factor – Not to exceed 0.003 for capacitance values of 100 pF or less: 0.001 for capacitance values greater than 100 pF.

Life: 150 percent dc rated voltage, at 125° +4,-0°C for 2,000 hours.
Group A test – 250 hours: group B continuation test for 1,750 hours.
For qualification test:
At 25°C:
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal or 0.5 pF, whichever is greater, from initial value.
Dissipation factor – Not to exceed 0.0045 for values of 100 pF, or less; 0.0025 for values greater than 100 pF.

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At 125°C:

Insulation resistance – Not less than 10,000 megohms.

Capacitance – Change not more than 0.5 percent nominal or 0.5 pF, whichever is greater, from initial 125°C value.

Dissipation factor – Not to exceed 0.009 for values of 100 pF or less; 0.007 for values greater than 100 pF.

For performance check:

Insulation resistance – Not less than initial requirement.

Capacitance – Change not more than 0.5 percent of nominal or 0.5 pF, whichever is greater, from initial value.

Dissipation factor – Not to exceed 0.0045 for values of 100 pF or less; 0.0025 for values greater than 100 pF.

For continuation test:

Capacitors shall meet requirements specified for qualification test.

Temperature coefficient: 140 ±25 parts/million/°C.

Capacitance drift: 0.1 percent or 0.1 pF, whichever is greater.

QUALITY ASSURANCE PROVISIONS

Sampling and inspection: MIL-C-11272 and as specified herein.

Qualification inspection: MIL-C-11272 on the respective non-TX version. (The TX version shall be procured from sources listed in QPL-11272 for the respective non-TX version.)

Quality conformance inspection: MIL-C-11272 and as follows:

Group A inspection: For the non-TX version, MIL-C-11272. For the TX version, 100 percent of the capacitors in a lot shall be subjected to the initial tests, power-stress conditioning, and post-stress conditioning measurements listed below. Visual and mechanical examinations shall be performed to 1 percent major and 2.5 percent minor AQLs. Successful performance of the barometric pressure and life tests on the respective non-TX version will suffice to demonstrate conformance to the requirement for TX version.

Group B inspection: MIL-C-11272: Successful performance of group B inspection for the respective non-TX version will suffice to demonstrate conformance to the requirement for TX version.

Initial tests: IR, capacitance, and DF shall be measured. Items that are out of capacitance tolerance, DF, and IR limits shall be removed from the lot.

Power-stress conditioning: Each capacitor intended for delivery as a TX part shall be subjected continuously to five times the rated voltage for 50 hours at an ambient temperature of 25°C.

Post power-stress conditioning measurement: After conditioning, each capacitor shall be measured for IR, capacitance, and DF. Items that fail to meet the initial requirements shall be removed from the inspection lot, and the quantity removed noted on the permanent lot history. Only lots having not more than 5 percent rejects or one capacitor, whichever is greater, as a result of this test shall be approved for delivery.

Acceptance: Acceptance of TX parts shall be based on the qualification, quality conformance inspection, sampling criteria of MIL-C-11272, and power-stress conditioning.

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Certification: Each lot delivered as TX parts shall be accompanied by a certificate verifying that power-stress conditioning has been performed and satisfactorily met.

Traceability: Each lot intended for delivery as TX parts shall be assigned a unique lot number prior to any tests. The lot number shall identify its date of manufacture. The manufacturer shall maintain a chronological file of these lot numbers and the data records shall be made available on request of the procuring activity. The data records shall be retained by the supplier for a period of 3 years.

Qualification: MIL-C-11272. Information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45401.

Marking: Non-TX parts in accordance with MIL-C-11272. TX parts shall be marked as follows:

Example:

CY06C
1R0C
TX

Custodian:
Air Force – 11

Review activities:
Air Force – 17, 19, 85

Preparing activity:
Air Force – 11

Agent:
DSA – ES

(Project 5910-F160)

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	☐ OTHER *(Specify)*: _____

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